

מגישה: שגית מלכה

1. רחוב הר מירון 2 הרצליה
2. וינגייט 182 הרצליה
3. משה שרת 3 הרצליה

א. עלויות נסיעה של כל אחד ושל כל תת קבוצה

מי נוסע	זמן (דקות)	מרחק (ק"מ)	תעריף
1	35	45.8	181.15
2	35	45.3	180.14
3	35	47.8	185.17
1 ל 2	15	4.3	57.53
1 ל 3	19	7	71
2 ל 3	6	2.4	35.62

2. חישוב ערך שאפלי והשוואה לתוצאות המתקבלות באלגוריתם הכללי

[הקוד כאן](#)

התוצאות יצאו זהות:

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## Ride-sharing homework example - 3 travelers

#### Inefficient calculation
Looping over all permutations of ['a', 'b', 'c']
Permutation ('a', 'b', 'c'):
  Player a: 181
  Player b: 57
  Player c: 35
Permutation ('a', 'c', 'b'):
  Player a: 181
  Player c: 71
  Player b: 22
Permutation ('b', 'a', 'c'):
  Player b: 180
  Player a: 58
  Player c: 35
Permutation ('b', 'c', 'a'):
  Player b: 180
  Player c: 35
  Player a: 58
Permutation ('c', 'a', 'b'):
  Player c: 185
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    Player a: 66
    Player b: 22
    Permutation ('c', 'b', 'a'):
    Player c: 185
    Player b: 30
    Player a: 58
    Shapley value of a is 100.81666666666668
    Shapley value of b is 82.11666666666667
    Shapley value of c is 91.36666666666666

#### Efficient calculation
Calculate Shapley-values for sub-problem with only  $d[0,1]$  (=
181.150000):
    Player 1 adds  $d[0,1]$  whenever he is first among 1,...,1, which happens
in 1/1 of the orders.
Calculate Shapley-values for sub-problem with only  $d[0,2]$  (=
180.140000):
    Player 2 adds  $d[0,2]$  whenever he is first among 1,...,2, which happens
in 1/2 of the orders.
    Player 1 removes  $d[0,2]$  whenever he is first among 1,...,2-1 that
comes after 2.
Calculate Shapley-values for sub-problem with only  $d[1,2]$  (= 57.530000):
    Player 1 adds  $d[1,2]$  whenever 1 is second and 2 is first among
1,...,2, which happens in  $1/(2-1+1)/(2-1)$  of the orders.
    Player 2 adds  $d[1,2]$  whenever 1 is second and 2 is first among
1,...,2, which happens in  $1/(2-1+1)/(2-1)$  of the orders.
Calculate Shapley-values for sub-problem with only  $d[0,3]$  (=
185.170000):
    Player 3 adds  $d[0,3]$  whenever he is first among 1,...,3, which happens
in 1/3 of the orders.
    Player 1 removes  $d[0,3]$  whenever he is first among 1,...,3-1 that
comes after 3.
    Player 2 removes  $d[0,3]$  whenever he is first among 1,...,3-1 that
comes after 3.
Calculate Shapley-values for sub-problem with only  $d[1,3]$  (= 71.000000):
    Player 1 adds  $d[1,3]$  whenever 1 is second and 3 is first among
1,...,3, which happens in  $1/(3-1+1)/(3-1)$  of the orders.
    Player 3 adds  $d[1,3]$  whenever 1 is second and 3 is first among
1,...,3, which happens in  $1/(3-1+1)/(3-1)$  of the orders.
    Player 2 removes  $d[1,3]$  whenever 1,3 join first and he joins third
among 1,...,3.
Calculate Shapley-values for sub-problem with only  $d[2,3]$  (= 35.620000):
    Player 2 adds  $d[2,3]$  whenever 2 is second and 3 is first among
2,...,3, which happens in  $1/(3-2+1)/(3-2)$  of the orders.
    Player 3 adds  $d[2,3]$  whenever 2 is second and 3 is first among
2,...,3, which happens in  $1/(3-2+1)/(3-2)$  of the orders.

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Shapley value of a is 100.81666666666668  
Shapley value of b is 82.11666666666666  
Shapley value of c is 91.36666666666666
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